FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VII

JEE (Main)-2023

TEST DATE: 22-03-2023

ANSWERS, HINTS & SOLUTIONS

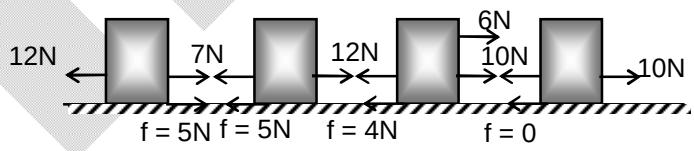
Physics

PART – A

SECTION – A

1. C

Sol. Acceleration = 0



2. D

Sol. $0 = \frac{1}{2}kx_0^2 - \mu mgx_0$

$\frac{kx_0}{2} = \mu mg$ …(i)

$K.E. = \frac{1}{2}k\left[x_0^2 - x^2\right] - \mu mg(x_0 - x)$

$\frac{d_{K.E.}}{dx} = \frac{1}{2}k\left[0 - 2x\right] - \mu mg(0 - 1)$

$kx = \mu mg$ …(ii)

$\Rightarrow x = \frac{x_0}{2}$

$K.E._{max} = \frac{1}{2}k\left[x_0^2 - \frac{x_0^2}{4}\right] - \mu mg\frac{x_0}{2}$

$= \frac{3}{8}kx_0^2 - \frac{kx_0^2}{4} = \frac{kx_0^2}{8}$

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3. A

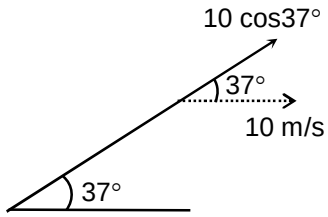
Sol. $\vec{v} = \vec{u} + \vec{a}t$

$0 = 10 \times \frac{4}{5} - 10 \times \frac{3}{5} t'$

$t' = \frac{4}{3}\text{sec}$

Time for maximum height from ground

$t = \frac{T}{2} + t' = 1 + \frac{4}{3} = \frac{7}{3}\text{sec}$



4. D

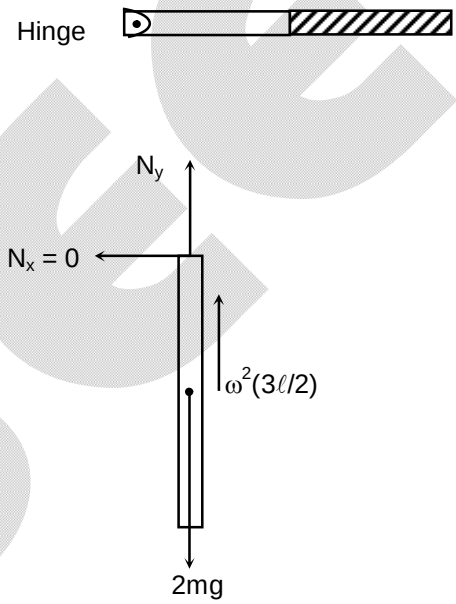
Sol. Conservation of energy $mg\frac{\ell}{2} + 2mg\frac{3\ell}{2} = \frac{5}{2}m\omega^2\ell^2$

$\frac{7}{2}mg\ell = \frac{5}{2}m\omega^2\ell^2$

$\frac{7g}{5\ell} = \omega^2$

$N_y - 2mg = 2m\omega^2\frac{3\ell}{2}$

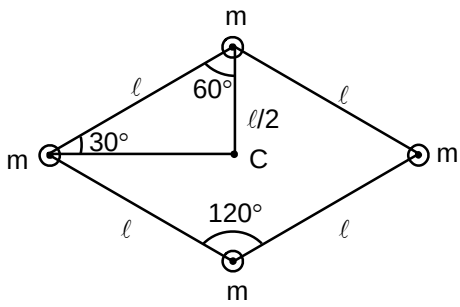
$N_y = \frac{31}{5}mg$



5. C

Sol. $\frac{1}{2}m_0v^2 = \frac{2Gmm_0}{\left(\frac{\ell}{2}\right)} + \frac{2Gmm_0}{\sqrt{3}\frac{\ell}{2}}$

$= \sqrt{\frac{8Gm}{\ell} + \frac{8Gm}{\sqrt{3}\ell}} = \sqrt{\frac{8Gm}{\ell}\left(1 + \frac{1}{\sqrt{3}}\right)}$



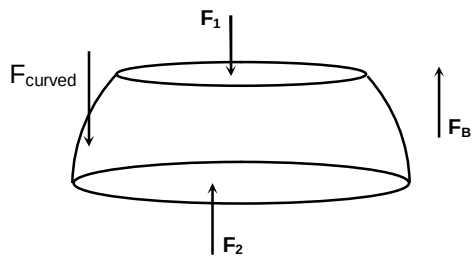
6. A

Sol. Buoyant force

$F_B = \frac{11}{24}\pi R^3\rho g$

$(P_0 + \rho gR)\pi R^2 - \left(P_0 + \rho g\frac{R}{2}\right)\pi\left(\frac{\sqrt{3}R}{2}\right)^2 - F_{curved} = F_B$

$F_{curved} = P_0\left(\frac{\pi R^2}{4}\right) + \frac{1}{6}\rho g\pi R^3$



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7. B

Sol. $y = A - A\cos\omega t$

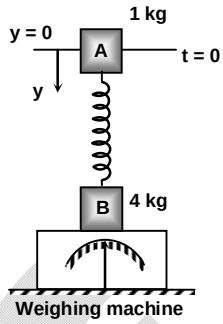
$= A[1-\cos\omega t]$

$y = \frac{m_A g}{k}\left[1-\cos\omega t\right]$

$N_B = M_B g + ky$

$= M_B g + M_A g\left[1-\cos\omega t\right] = 40 + 10\left[1-\cos\omega t\right]$

$N_B = 50 - 10\cos\omega t$



8. D

Sol. $U_i + W_{Battery} = U_f + Heat$

$\frac{CV^2}{3} + \frac{4CV^2}{3} = CV^2 + Heat$

$Heat = \left(\frac{5}{3}-1\right)CV^2 = \frac{2}{3}CV^2$

9. C

Sol. $\rho = 4\rho_0 x, \quad 0 \le x \le \frac{1}{4}$

$\rho = \rho_0, \quad \frac{1}{4} \le x \le 0$

$q_{enc} = \int_0^{1/4} 4\rho_0 x dx + \frac{3}{4}\rho_0 \times 1$

$= \frac{4\rho_0}{2}\times\frac{1}{16} + \frac{3\rho_0}{4} = \frac{\rho_0}{4}\left[\frac{1}{2}+3\right]$

$\frac{q_{enc}}{\varepsilon_0} = \frac{7\rho_0}{8\varepsilon_0}$

$\phi = \frac{7}{8}$

10. B

Sol. Current through inductor, $I = \frac{E}{2R}\left[1-e^{-\frac{2Rt}{3L}}\right]$

$\frac{dI}{dt} = \frac{E}{2R}\left[0+\frac{2R}{3L}e^{-\frac{2Rt}{3L}}\right]$

$\frac{dI}{dt} = \frac{E}{3L}e^{-\frac{2Rt}{3L}}$

$I' = \frac{L\frac{dI}{dt}}{R} = \frac{Ee^{-\frac{2Rt}{3L}}}{3R}$

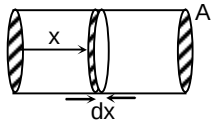
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11. A

Sol. $\text{Heat} = \left(\frac{B\ell v}{R}\right)^2 \times R \times 2 = \frac{2B^2\ell^2v^2}{R} = \frac{2\times4\times1\times1}{2}$

12. C

Sol. $\int dR = \int_0^{\ell} \frac{x}{\sigma_0 \ell}\frac{dx}{A} = \frac{\ell}{2\sigma_0 A}$



Current

$I = \frac{V}{R} = \frac{V2\sigma_0 A}{\ell}$

$\frac{I}{A} = J = \frac{2v\sigma_0}{\ell}$

$\sigma E = \frac{2v\sigma_0}{\ell}$

$\frac{\sigma_0 \ell E}{x} = \frac{2v\sigma_0}{\ell}$

$E = \frac{2vx}{\ell^2}$

13. B

Sol. Modulation index

$\mu = \frac{A_m}{A_C} = \frac{5}{10} = 0.5$

$\text{Band width} = 2f_m = \frac{2\times2000\pi}{2\pi} = 2000\text{Hz}$

14. C

Sol. $m_{\perp} = (-)\frac{v}{u} = \frac{f}{f-u} = \frac{10}{10-(-20)}$

$m_{\perp} = \frac{1}{3}$

$\vec{v}_{IM} = -m_{\perp}^2 \vec{v}_{OM}$

$\vec{v}_{IM} = -\left(\frac{1}{9}\right)8$

$\vec{v}_{IG} = \vec{v}_{IM} + \vec{v}_{MG} = -\frac{8}{9} + 2 = \frac{10}{9}$ cm/s

15. C

Sol. $\phi_A = 13.6$ eV; $\phi_B = 7.2$ eV

$\text{K.E.}_B = 2\text{K.E.}_A$

$\frac{12400}{\lambda} - 7.2 = 2\left(\frac{12400}{\lambda} - 13.6\right)$

$27.2 - 7.2 = \frac{12400}{\lambda}$

$20 = \frac{12400}{\lambda}$

$\lambda = 620\text{Å}$

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16. B

Sol. $\tau = \frac{t_0}{2}$

$\tau = \frac{0.693}{\lambda} = \frac{t_0}{2}$

$\lambda = \frac{2\ell n2}{t_0}$

$\frac{2N_0}{3} = N_0 e^{-\lambda t_1}$

$\frac{N_0}{3} = N_0 e^{-\lambda t_2}$

$2 = e^{\lambda(t_2 - t_1)}$

$\ell n2 = \lambda(t_2 - t_1)$

$\ell n2 = \frac{2\ell n2}{t_0}(t_2 - t_1)$

$\frac{t_0}{2} = (t_2 - t_1)$

$(t_2 - t_1) = 20$

17. A

Sol. Radius of gyration, $R \propto G^a h^b c^c$

$\left[M^0LT^0\right] \propto \left[M^{-1}L^3T^{-2}\right]^a \left[ML^2T^{-1}\right]^b \left[LT^{-1}\right]^c$

Comparing powers, a = 1/2, b = 1/2, c = –3/2

18. B

Sol.

A	B	Y
1	0	1
0	1	1
1	1	0
0	0	1

19. B

Sol. Least count = 1 M.S.D – 1 V.S.D

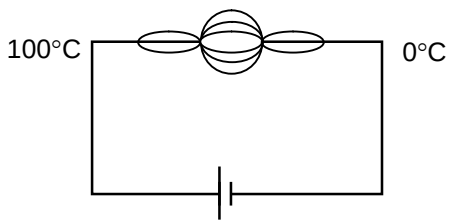
= 0.05 – 0.047 = 0.003

Reading = 6.10 + 24 × 0.003 = 6.172

20. B

Sol. $100 - \frac{100}{\left(\frac{5R}{6}\right)} \times \frac{R}{3} = 60°$

(where R is thermal resistance of each rod)



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SECTION – B

21. 4

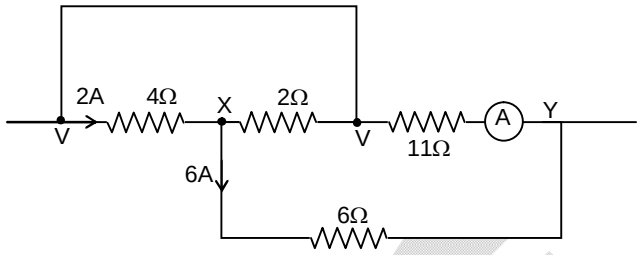
Sol. $\frac{V-x}{4}=2$

$V-x=8$

$x-y=36$

$y=x-36$

$I=\frac{V-y}{11}$



22. 12

Sol. Let x be the amount of water and y be the amount of ice at the end of second minute

$yL_{ice}+(x+y)C_{water}\Delta T=30\times 60$

$400y+(x+y)4\times 2=1800$ …(i)

$(x+y)\times 4\times 6=30\times 60$ …(ii)

Solving (i) and (ii)

$400y+\frac{1800}{6}\times 2=1800$

$400y=1200$

$\Rightarrow y=3g$

Initial amount of ice = $\frac{30\times 120}{400}+3=12g$

23. 140

Sol. $W=nR(400)\ell n\left(\frac{P_B}{P_C}\right)+nR(300)\ell n\left(\frac{P_D}{P_A}\right)$

24. 10

Sol. $\frac{1}{f_e}=\frac{1}{v_e}-\frac{1}{u_e}$

$\frac{1}{6.25}=\frac{1}{-25}-\frac{1}{u_e}$

$\frac{1}{u_e}=-\frac{1}{25}-\frac{1}{6.25}$

$u_e=-5$

$|v_0|+|u_e|=15$

$v_0=10$ cm

$\frac{1}{f_0}=\frac{1}{v_0}-\frac{1}{u_0}$

$\frac{1}{+5}=\frac{1}{+10}-\frac{1}{u_0}$

$u_0=-10$

25. 16

Sol. $\frac{I_0}{2}\left(\frac{3}{4}\right)^4\times\frac{100}{I_0}=15.82\%=16\%$

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26. 8

Sol. $\vec{P} = P\hat{i} + P\hat{j}$

$\vec{r}_A = 3i - 4\hat{j}$

$\vec{r}_B = 3i + 4\hat{j}$

$V_A = \dfrac{K\vec{P}\cdot\vec{r}_A}{r_A^3} = \dfrac{-KP}{125}$

$V_B = \dfrac{K\vec{P}\cdot\vec{r}_B}{r_B^3} = \dfrac{7KP}{125}$

Work done = $q_0(V_B - V_A) = \dfrac{8}{125}(KPq_0) = 8$ Joule

27. 6

Sol. Intensity, $\dfrac{\text{Power}}{\text{Area}} = \dfrac{1}{2}\varepsilon_0 E_0^2 C$

$\dfrac{2.4}{100} \times \dfrac{100}{4\pi(2)^2} = \dfrac{1}{2}\varepsilon_0 E_0^2 C$

$\Rightarrow E = 6$ V/m

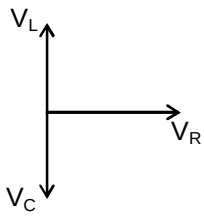
28. 8

Sol. $B\pi R^2 I \geq Mg\dfrac{d}{2}$

$I \geq \dfrac{Mgd}{2\pi R^2 B}$

29. 5

Sol. $\cos\phi = \dfrac{V_R}{V} = \dfrac{50}{130} = \dfrac{5}{13}$



$V_R^2 + V_L^2 = (100)^2$

$130 = \sqrt{V_R^2 + (V_L - V_C)^2}$

$16900 = V_R^2 + 14400$

$V_R = \sqrt{2500} = 50$ V

30. 2

Sol. Apply Bernoulli's between B and C

$P_B + \dfrac{1}{2}\rho V_B^2 = P_C + \dfrac{1}{2}\rho V_C^2$ …(i)

Also, $V_C = 2V_B$ …(ii)

$V_B = \sqrt{\dfrac{2g}{3}\Delta h_{BC}}$

Applying Bernoulli's between A and B

$P_A + \dfrac{1}{2}\rho V_A^2 = P_B + \dfrac{1}{2}\rho V_B^2$ …(iii)

$V_A = \sqrt{2g\left[\dfrac{\Delta h_{BC}}{3} - \Delta h_{AB}\right]}$

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Chemistry

PART – B

SECTION – A

31. C

Sol.

Molecule/ion	Number of unpaired electron	Bond order	Magnetic character
O_2^{-2}	1	1.5	Para
O_2^{+}	1	2.5	Para
C_2^{-2}	0	3	Dia
N_2^{-4}	0	1	Dia

$$\text{Bond length} \propto \frac{1}{\text{Bond order}}$$

32. B

Sol. $P(V-b)=RT$

$PV-bP=RT$

Divide by RT

$$\frac{PV}{RT}-\frac{bP}{RT}=1$$

$$Z=1+\left(\frac{b}{RT}\right)P$$

Slope $\propto$ b $\propto$ size of atom.

33. A

Sol. $Na(g)+e\longrightarrow Na^{-}(g)$ $\Delta H=-53$ kJ / mole

Rest B, C and D are endothermic.

34. D

Sol. A, B and C are correct options.

35. D

Sol. $r_n=r_o\frac{n^2}{Z}$

$$\Delta R=r_{n_2}-r_{n_1}=\frac{r_o}{Z}\left(n_2^2-n_1^2\right)$$

$$He^{+},\ \Delta R_1=\frac{r_o}{2}\times\left(3^2-2^2\right) \quad \ldots (i)$$

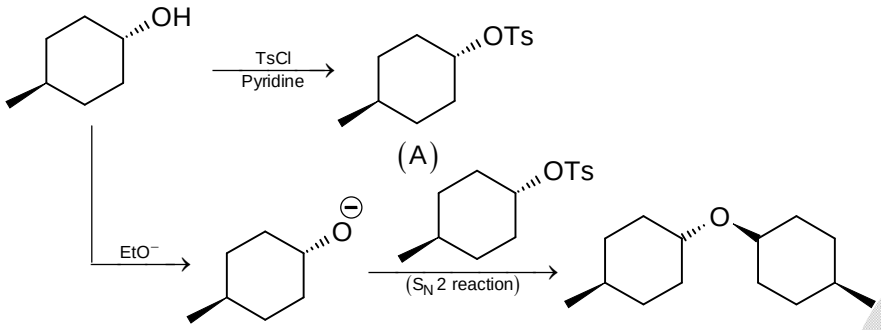
$$Li^{+2},\ \Delta R_2=\frac{r_o}{3}\times\left(4^2-3^2\right) \quad \ldots (ii)$$

$$\frac{\Delta R_1}{\Delta R_2}=\frac{3}{2}\times\frac{5}{7}=\frac{15}{14}$$

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36. C

Sol.



37. D

Sol. Biuret test detects presence of peptide bonds.

- Barfoed's test based on reduction of copper (II) acetate to copper (I) oxide which forms brick red precipitate.
- Molisch's test is for presence of carbohydrate.

38. D

Sol. Glyptal is used to manufacture of paints and lacquers.

39. C

Sol. Due to deficiency of Vitamin E.

40. C

Sol. B.H.T. and BHA are most familiar anti oxidants.

41. A

Sol. CdS, ZnO, Graphite are the examples of hexagonal.

42. C

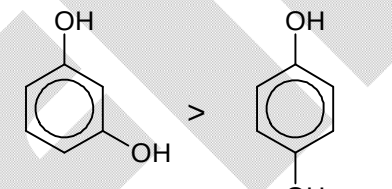
Sol. $AB \rightleftharpoons A^+ + B^-$

$K_{sp} = \left[A^+\right]\left[B^-\right] = 5\times10^{-9}\times10\times10^{-9}$

$= 50\times10^{-18} = 5\times10^{-17}\ M^2$

43. D

Sol.



(pK_a = 9.44)

>



(pK_a = 9.96)

(Correct acidic strength)

44. C

Sol. Reverbertory furnace is used in sulphides of copper.
A, B and D are correct statements.

45. D

Sol. A – B has highest bond enthalpy so stiffest bond.
A – C has the largest bond length.

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46. D

Sol. Let us assume reactivity of C – D is a and C – H is b.

$$\frac{2a}{2a+6b}=\frac{50}{100}$$

$$\frac{a}{a+3b}=0.5 \Rightarrow a=0.5a+1.5b$$

$$0.5a=1.5b \Rightarrow \frac{a}{b}=\frac{3}{1}$$

47. B

Sol. Wave length range is 200 – 315 nm (UV region)

48. D

Sol. A, B and C are correct statements. Repeated used of DDT in insects is not good. So, Aldrin and Dieldrin were introduced in place of DDT in the market.

49. C

Sol. In option A, B and D entropy is increasing where as in C option when pressure is increased O_2 molecules come closer to each other hence entropy decreases $\Delta S < 0$.

50. C

Sol. $\Delta G^{o} = -2.303\ RT \log_{10} K_{eq.}$

$-413.15 = -2.303 \times 2 \times 298 \log_{10} K_{eq}$

$= 0.3010 = \log_{10} K_{eq}$

$K_{eq} = 2$

Hence, graph (C) matches correctly.

SECTION – B

51. 5

Sol.

	$3A \longrightarrow B$	
$t = 0$	a	0
$t = 240$ hr	$a - b$	$b/3$

$$a-b=\frac{b}{3} \Rightarrow a=\frac{3b}{4} \Rightarrow b=\frac{3a}{4}$$

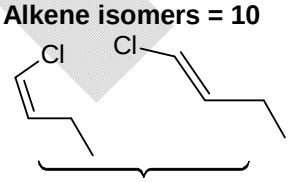
$\therefore$ a to $\frac{a}{4}$ time given is 240 hours which is equal to $2t_{1/2}$ for A.

Hence, $2t_{1/2} = 240 \Rightarrow t_{1/2} = 120$ hrs $= 5$ days

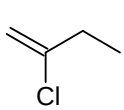
52. 4

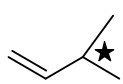
Sol. Degree of unsaturation of C_4H_7Cl is 1 hence alkenes and cyclo compounds are possible.

Alkene isomers = 10

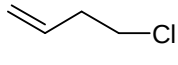


cis-trans

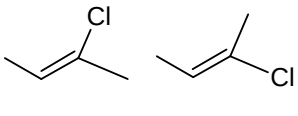




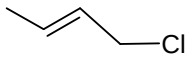
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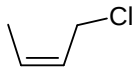
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trans-cis

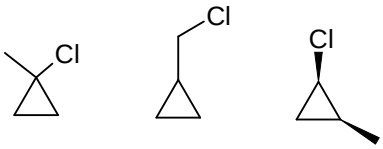


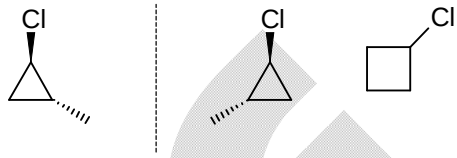
trans



cis

Cyclic isomers = 6





$x - y = 10 - 6 = 4$

53. 4

Sol. Only CH_3CHO will react with Fehling solution.

$$CH_3CHO + 2Cu^{+2} \longrightarrow CH_3COO^- + Cu_2O(\downarrow)$$

44 (F.S.) 143 Red

$\because$ 143 gm of Cu_2O is formed by = 44 gm CH_3CHO

$\therefore$ 0.975 gm of Cu_2O is formed by $= \frac{44}{143} \times 0.975$

0.3 gm of CH_3CHO

Mass of acetone = 0.2 gm

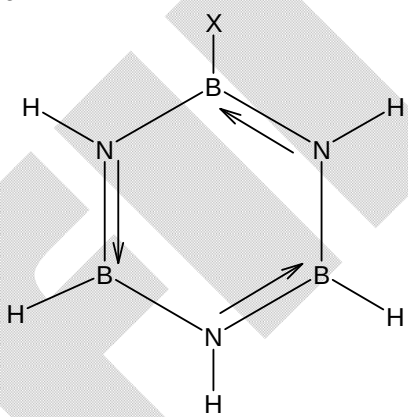
% of acetone $= \frac{0.2}{0.5} \times 100 = 40\%$

x = 40

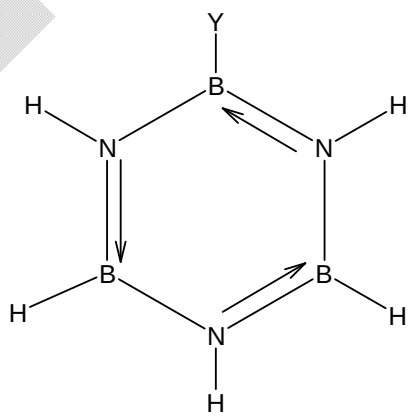
So, $\frac{x}{10} = 4$

54. 6

Sol.



(Y has 3 different positions)



(X has 3 different positions)

Total position isomers = 3 + 3 = 6.

55. 6

Sol. $NH_2CONH_2 + 2NaOH \longrightarrow 2NH_3 + Na_2CO_3$

Moles of NH_3 = moles of HCl $= \frac{100 \times 2}{1000}$ = 0.2 moles

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Moles of $NH_2CONH_2 = \frac{1}{2}$ moles of $NH_3 = \frac{1}{2} \times 0.2 = 0.1$

Mass of urea $= 0.1 \times 60 = 6$ gm

56. 5

Sol. $\frac{P^o - P_s}{P^o} = \frac{n}{N} = \frac{n_A}{n_B} = P_s = \frac{9}{10}P^o$

$$\frac{P^o - \frac{9}{10}P^o}{P^o} = \frac{W_A}{W_B} \times \frac{M_B}{M_A}$$

$$\frac{1}{10} = \frac{a}{b} \times \frac{1}{2} \Rightarrow \frac{b}{a} = 5$$

57. 3

Sol. $AB_2(g) \longrightarrow A^+(g) + B^-(g) + B(g)$

$\Delta H_r = 2\Delta H_{A-B} + I.E._{(A)} + \Delta Heg(B)$

$= 2 \times 100 + 250 - 150$

$= 200 + 250 - 150$

= 300 kcal/mole.

So, x is 300 hence $\frac{x}{100}$ will be 3.

58. 6

Sol. $2NaClO_3 \xrightarrow{\Delta} 2NaCl + 3O_2$

$NaCl + AgNO_3 \longrightarrow AgCl + NaNO_3$

Moles of $O_2 = \frac{2}{3}$ moles of NaCl $= \frac{2}{3}$ moles of AgCl

Moles of AgCl $= \frac{2}{3}$ moles of $O_2 = \frac{2}{3} \times \frac{2}{32}$

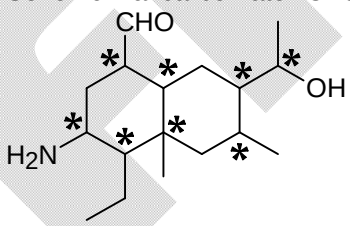
Mass of AgCl $= \frac{2}{3} \times \frac{2}{32} \times 143.5$ gm

= 5.97

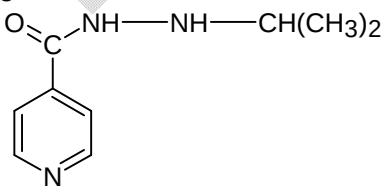
$\simeq 6$

59. 7

Sol. Seven chiral carbon atoms have hydrogen.



60. 3

Sol. 

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Mathematics

PART – C

SECTION – A

61. C

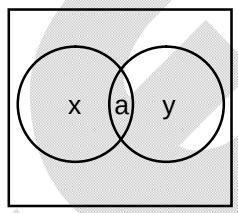
Sol. For one to 6, 3 black and 3 white and 7th ball is white = $^6C_3\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^3\cdot\frac{1}{2}=\frac{20}{128}=\frac{5}{32}$

62. C

Sol. $(x+y)\geq(x+y+a)(1-(x+y+a))$

$\Rightarrow x+y\geq x+y+a-(x+y+a)^2$

$\Rightarrow (x+y+a)^2\geq a$



63. A

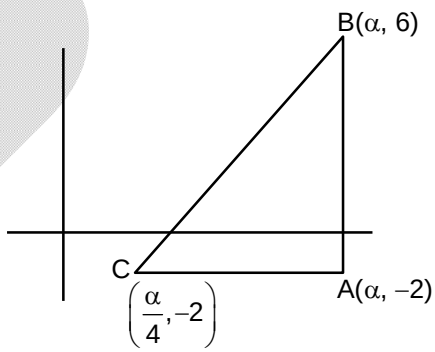
Sol. $8\cos x\left(\cos^2 x-\sin^2\frac{\pi}{6}-\frac{1}{2}\right)=1$

64. D

Sol. It is differentiable at x = 0 also differentiable at $x=\pi$

65. B

Sol. Circum centre is mid point of AB



66. C

Sol. Let $I = I_n + I_4 = \int(\tan^4x+\tan^6x)dx = \int\tan^4x+\sec^2x+dx = \frac{\tan^5 x}{5}+c$

$\Rightarrow a=\frac{1}{5}$, b = 0

67. C

Sol. The normal vector is $\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&-2&3\\2&-1&-1\end{vmatrix}=5\hat{i}+7\hat{j}+3\hat{k}$

Now plane is 5x + 7y + 3z + 5= 0 distance from (1, 1, 1) = $\left|\frac{5+7+3+5}{\sqrt{25+49+9}}\right|=\frac{20}{\sqrt{83}}$

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68. A

Sol. $5(\sec^2 x - 1) = 9\cos^2 x + 7$

Let $\cos^2 x = t$

$= t = \frac{1}{3}, t \neq -\frac{5}{3}$

69. A

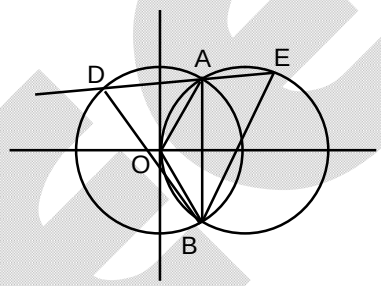
Sol. $I_n = (n-1)(n-w)(n-w^2)$

$I_n = n^3 - 1$

$S_n = \sum n^3 - n$

70. A

Sol. See the figure $\angle AOB = 120°$



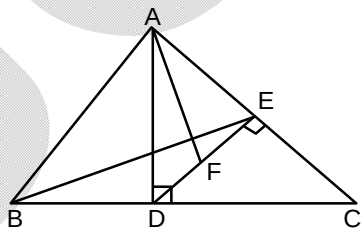
71. A

Sol. Let $D \equiv (0, 0)$

$B \equiv (-b, 0)$

$C \equiv (b, 0)$

$A\ (0, a)$



72. B

Sol. The mid point lie on $x + y = 0$

Find the equation of the chord by $T = S_1$ and by taking mid point $(-t, t)$

73. A

Sol. Let $\vec{R} = x\hat{i} + y\hat{j} + z\hat{k}$

$\vec{R} \times \vec{B} = \vec{C} \times \vec{B}$

$$\Rightarrow \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ 1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -3 & -7 \\ 1 & 1 & 1 \end{vmatrix}$$

74. D

Sol. $T_n = n^3(n+1)$

$T_n = n^4 + n^3$

75. C

Sol. $$f(x) = \frac{\sin\left(\frac{x}{2}\right)}{\sqrt{\sec^2\left(\frac{x}{2}\right)}} - \frac{\cos\left(\frac{x}{2}\right)}{\sqrt{\operatorname{cosec}^2\left(\frac{x}{2}\right)}}$$

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$$f(x)=\frac{\sin\left(\frac{x}{2}\right)}{\left|\sec\frac{x}{2}\right|}-\frac{\cos\left(\frac{x}{2}\right)}{\left|\operatorname{cosec}\frac{x}{2}\right|}$$

76. D

Sol. Put $f^{-1}(x) = t$

$$I=\int_2^1\frac{f'(t)dt}{t^5+5t}=\ln 7$$

77. A

Sol. Put $x = t^2 = dx = 2t\,dt$

$$I=\int_0^1\frac{t(2t)dt}{\left(t^2+1\right)\left(1+3t^2\right)\left(3+t^2\right)}$$

78. B

Sol. Put $x = 1, y = 1 \Rightarrow f(1) = 0$

Diff. w.r.t (x)

$yf'(xy) = f(y) + yf'(x)$

Put $x = 1$, $yf'(y) = f(y) + y$

79. D

Sol. Let $x - 1 = X$

$x - 1 = 0 \equiv x = 0$

$y^2 = 5 - x \equiv y^2 = 4 - X$ ...(1)

Let the circle is $(X - r)^2 + (y - 0)^2 = r^2$ ...(2)

From (1) and (2)

$X^2 - (2r + 1)x + 4 = 0$

Now $D = 0 \Rightarrow r = 3/2$

80. B

Sol. Point of intersection of tangents are $(at_1t_3, a(t_1 + t_3))$

From given we can conclude t_1 and t_3 both satisfying $at^2t_2^2 = 4b(t + t_2)$

$\Rightarrow at^2t_2^2 - 4bt - 4bt_2$

$$t_1+t_3=\frac{4b}{at_2^2},\ t_1t_3=\frac{4bt_2}{at_2^2}$$

SECTION – B

81. 3

Sol. $2\sin x + \tan x \geq 3x$

Also equality holds when $x = 0$

82. 15

Sol. Here focus chord is normal. Hence the focal chord is the major axis.

83. 8046

Sol. Write general terms of $(1 + x^2)^6 (1 + x^3)^9 (1 + x^4)^{12}$

84. 17

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Sol. Write general terms = $^{128}C_r \left(\sqrt{2}\right)^{128-r} \left(\sqrt[8]{7}\right)^r$

for r = 8k, k = 0 to 16 are possible.

85. 0

Sol. $P^3 = Q^3$ and $P^2Q = Q^2P$

$\Rightarrow P^3 - P^2Q = Q^3 - Q^2P$

$\Rightarrow P^2(P - Q) = Q^2(Q - P)$

$\Rightarrow |P^2 (P - Q)| = |Q^2(Q - P)|$

$\Rightarrow |(P^2 + Q^2) (P - Q)| = 0$

86. 1

Sol. $a = \log_{3^2 \cdot 5}^{5^2 \cdot 3} = \dfrac{2\log_{10}^5 + \log_{10}^3}{2\log_{10}^3 + \log_{10}^5}$

$b = \log_{3^3 \cdot 5}^{5^3 \cdot 3} = \dfrac{3\log_{10}^5 + \log_{10}^3}{3\log_{10}^3 + \log_{10}^5}$

Put the values of a and b is ab + 5(a – b).

87. 4

Sol. Let the equation of normal is $y + tx = 4t + 2t^3$ and it passes through (3, 3)

$3 + 3t = 4t + 2t^2$

This equation has only are real value of t i.e. t = 1

so only are normal ordinate = 2at = 4

88. 22

Sol. $60 = 2 \times 2 \times 3 \times 5 = 2^2 \times 3 \times 5$

2^2 can be distributed = 5 ways

3 and 5 both 3 ways

89. 1

Sol. $|P'(1)| = \lim\limits_{h \to 0} \left| \dfrac{P(1+h) - P(1)}{h} \right|$

Also $|P(1)| \le 0 \Rightarrow P(1) = 0$

$\Rightarrow |P'(1)| = \lim\limits_{h \to 0} \left| \dfrac{P(1+h)}{h} \right| \le \lim\limits_{h \to 0} \left| \dfrac{e^h - 1}{h} \right|$

$\Rightarrow |P'(1)| \le 1$

90. 176

Sol. Apply vector triple product

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